

Kunli Liu, PhD

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SUMMARY

AI-native staff-caliber engineer specializing in large-scale distributed infrastructure and production LLM systems. Builds end-to-end agentic AI platforms—from transformer-based agent design and inference serving to evaluation frameworks and AI-driven automation—with measurable impact on developer velocity and revenue. Leads cross-org engineering initiatives and drives technical direction from research to production.

Languages: Hack/PHP, C++, Python, PyTorch, SQL, PySpark

AI/ML: transformer architecture, attention mechanisms, normalization techniques (LayerNorm, RMSNorm), BPE tokenization, pre-training, post-training alignment (SFT, RLHF, RLVR, DPO), scaling laws, CNNs, computer vision, feature engineering, distributed training (data/model/pipeline parallelism), GPU kernel optimization (CUDA)

AI Systems: Large Language Models (LLMs), Retrieval Augmented Generation (RAG), model serving & inference optimization, prompt engineering, model evals, LLM observability & tracing, AI-driven automation

Data & Databases: SQL, Apache Spark, distributed data pipelines, ETL, query optimization, data modeling, Pandas, NumPy

Distributed Systems: orchestration, phased rollout, sharding, observability, reliability, E2E integration testing, CI/CD

Engineering Leadership: multi-team roadmap planning, cross-org partner alignment, headcount & capacity planning

EDUCATION

University of Pennsylvania

Philadelphia, PA

Master of Science in Computer Science

Selected Coursework: Computer Vision, Database & Information Systems, Big Data Analytics

Michigan State University

East Lansing, MI

Doctor of Philosophy

EXPERIENCE

Senior Software Engineer, Meta Platforms

2022 – Present

Sunnyvale, CA

IC5 · Greatly Exceeds Expectations (top 10% performer)

Agentic AI Infrastructure for Ads Serving

- Architected and productionized a first-of-its-kind LLM systems with deep integration into the ads delivery stack, enabling autonomous reasoning over serving infrastructure telemetry at production scale.
- Built low-latency inference serving infrastructure for the agent—including context management, tool-use orchestration, multi-step reasoning pipelines, and investigation runbook automation—and led adoption across pre-impression delivery services, establishing the foundational LLM agent framework for ads infrastructure teams.
- Validated agent efficacy through structured evaluation benchmarks measuring production incident detection accuracy and mitigation quality, projecting *\$20M+* in annual revenue savings from by reducing time to detection and mitigation.

Unified E2E Test Platform — “TestPilot”

- Led 10+ engineers across ads infra, products, and change management to deliver a unified E2E testing framework spanning the full ad lifecycle on Facebook and Instagram: pre-delivery (publishing, indexing, matching, ranking) and post-delivery (impression, clicks, conversion, revenue).
- Owned TestPilot architecture; drove key design tradeoffs across deterministic vs. non-deterministic delivery and joint vs. isolated orchestration; designed LLM evaluation pipelines—scoring rubrics, benchmark suites, automated quality gates—to validate model-generated artifacts.
- Shipped **AI-TestPilot**, an agentic code-generation system using prompt-engineered LLMs for automated test authoring, validation, and enablement, driving ads targeting E2E test coverage from 0% to 95%.
- Launched TestPilot for pre-production full ad lifecycle validation, directly addressing *\$300M/year* in revenue exposure from reliability risks and product development inefficiencies, earning recognition from org directors and VPs.

Less Personalized Ads (LPA)

- Reliability area lead for LPA: built testing infrastructure from zero, where no prior reliability coverage existed, architected the full coverage model (canary, CI/CD integration, privacy API contracts).
- Drove launch-readiness decisions (gating criteria, failure modes, blast radius controls, exception handling), unblocking LPA launch and preventing *\$10M* revenue loss.
- Shipped observability platform for LPA covering 100+ delivery-critical metrics across pre-commit, deployment, and production stages; defined SLOs, dashboards, and alert thresholds to reduce incident risk and shorten detection/triage loops.

Software Engineer Intern, Oracle Corporation

05/2021 – 08/2021

Seattle, WA

- Built an event-driven E2E test automation pipeline for Oracle Hardware validation, triggering fleet-level validation on hardware events and eliminating manual test cycles for release qualification.

PUBLICATIONS

11 peer-reviewed publications (ACS, IEEE, Wiley, Elsevier, RSC) | **h-index 7** | **146 citations** ([Google Scholar](#)). Research applies **deep learning**, **computer vision**, and **machine learning** to intelligent biomedical systems: **neural network-based 3D depth estimation** for surgical AI guidance; **ML-enhanced spectral signal analysis** for intelligent sensing and imaging; **multi-modal imaging pipelines** for **AI-assisted cancer diagnostics**; and **ML-accelerated drug discovery** and nanomaterial design.

1. **K. Liu** et al., “Monocular depth estimation based on **deep learning** for intraoperative guidance,” *Photonics Research*, Optica/Chinese Laser Press (**Impact Factor 7.2**), 2025.
2. **K. Liu** et al., “Robust synthesis of targeting glyco-nanoparticles for surface enhanced resonance Raman scattering,” *Small Science*, Wiley (**Impact Factor 8.3**), 2024.
3. **K. Liu** et al., “Combinatorial synthesis of a hyaluronan based polysaccharide library,” *Carbohydrate Polymers*, Elsevier (**Impact Factor 12.5**), 2023.
4. **K. Liu** et al., “Synthesis of self-assembled hyaluronan based nanoparticles and their applications,” *Carbohydrate Research*, Elsevier (**Impact Factor 2.5**), 2022.
5. **K. Liu** et al., “MOEMS-enabled miniaturized SERS Raman detection system,” *MOEMS and Miniaturized Systems XXIV*, SPIE (Conf. Proc.), 2025.
6. **K. Liu** et al., “Active targeting hyaluronan conjugated nanoprobe for magnetic particle imaging,” *ACS Applied Materials & Interfaces*, American Chemical Society (**Impact Factor 8.3**), 2024.
7. **K. Liu** et al., “Cost-effective near infrared fluorescence wide-field camera,” *IEEE Photonics Technology Letters*, IEEE (**Impact Factor 2.8**), 2023.
8. **K. Liu** et al., “Indocyanine green-conjugated superparamagnetic iron oxide nanoworm,” *ACS Applied Nano Materials*, American Chemical Society (**Impact Factor 5.5**), 2022.
9. **K. Liu** et al., “Divergence of chemerin reduction by an ATS9R nanoparticle,” *Biomedicines*, MDPI (**Impact Factor 3.9**), 2022.
10. **K. Liu** et al., “Synthesis and immunological evaluation of the unnatural β -linked mucin-1,” *Organic & Biomolecular Chemistry*, Royal Society of Chemistry (**Impact Factor 2.8**), 2021.
11. **K. Liu** et al., “Exploration of human xylosyltransferase for chemoenzymatic synthesis,” *Organic & Biomolecular Chemistry*, Royal Society of Chemistry (**Impact Factor 2.8**), 2021.